

Relational operators [== (comparison) , < , > , <= , >= , != (not equal)]:

They are used to check the given condition or expression is true or false. If condition true always the answer is 1 and false it is 0.

```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 20 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",3==3);
printf("%d\n",9==7);
printf("%d\n",9==9.0);
printf("%d\n",0.90 == .9 );
printf("a addr %u\n","a");
printf("a addr %u\n","a");
printf("%d\n","a"=="a");
printf("%d\n",'a'==97);
printf("%d\n",'a'>'A'); /* 97 > 65 */
printf("%d\n",'a'<='A'+32);
printf("%d\n",7!=7);
printf("%d\n",7.01>7);
getch();
}
```

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F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

11:38 AM
06-Feb-25

```
1
0
1
1
1
a addr 431
a addr 444
0
1
1
1
1
0
1
1
```

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Operator precedence / Operator priority (ASSOCIATION OF OPERATORS)

1. ()
2. +, -, ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, /, %
5. +, - (Binary)

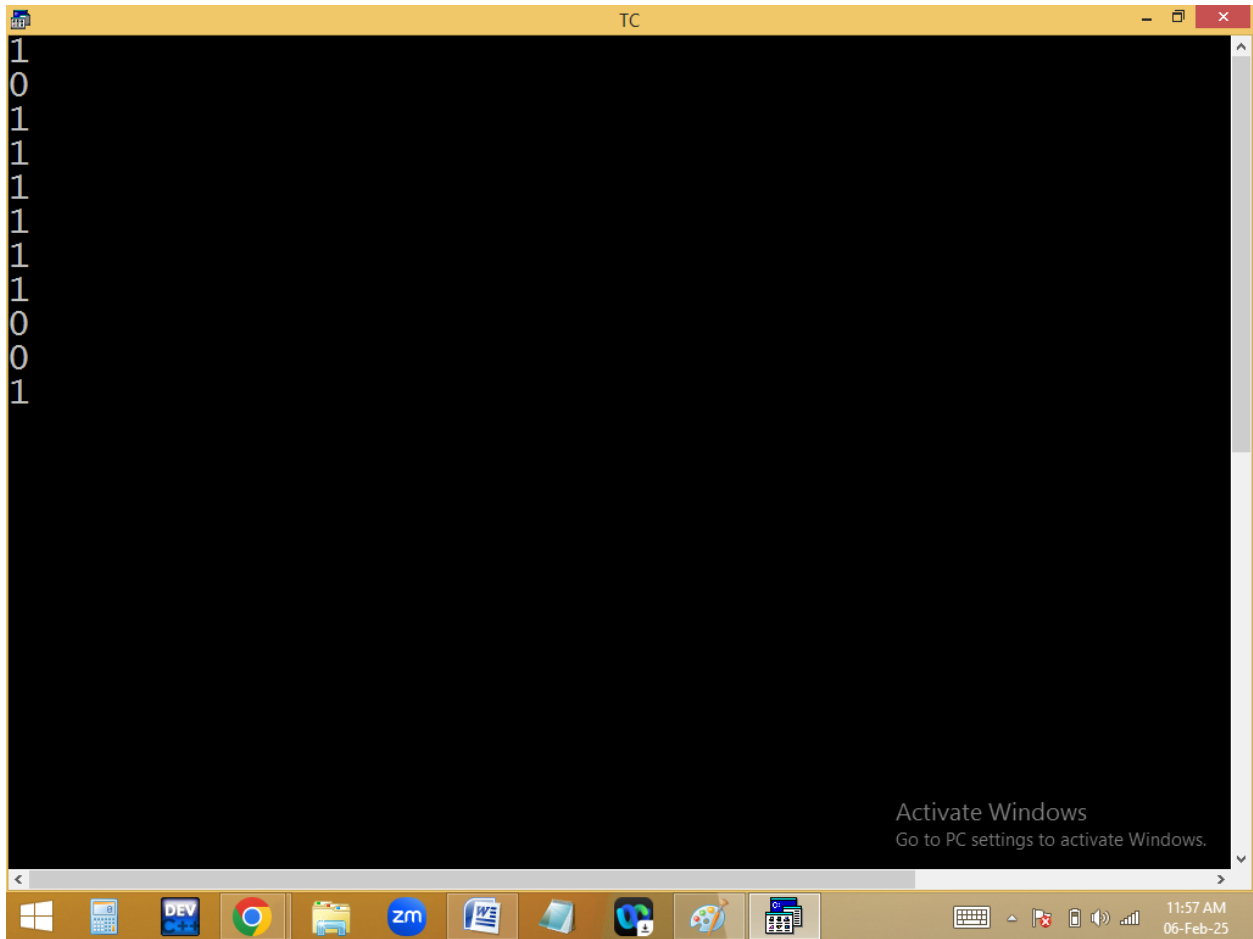
6. ==, !=
7. &&
8. ||
9. ?: (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)

```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 30 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", '0' > 0); /* 48 > 0 */
printf("%d\n", 5+3/2==4);
printf("%d\n", (5+3)/2==4);
printf("%d\n", 1+2-2==1);
printf("%d\n", 5*3/2==7);
printf("%d\n", 5/3*2==2);
printf("%d\n", 5-3+2==4);
printf("%d\n", 2+3*4+5==19);
printf("%d\n", 2+3*4+5==45);
printf("%d\n", 2+3*4+5==25);
printf("%d\n", (2+3)*(4+5)==45);
getch();
}
```

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```
TC
File Edit Run Compile Project Options Debug
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5=5);
getch();
}
```

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5 == 5 = 5

↓

↓

1 = 5

↑

constant ==> Error

Logical operators:

1. && - Logical and
2. || - Logical or
3. ! – Logical not

&&, || used to combine two or more expressions into a single expression.

Not used for negation i.e. true becomes false and false become true.

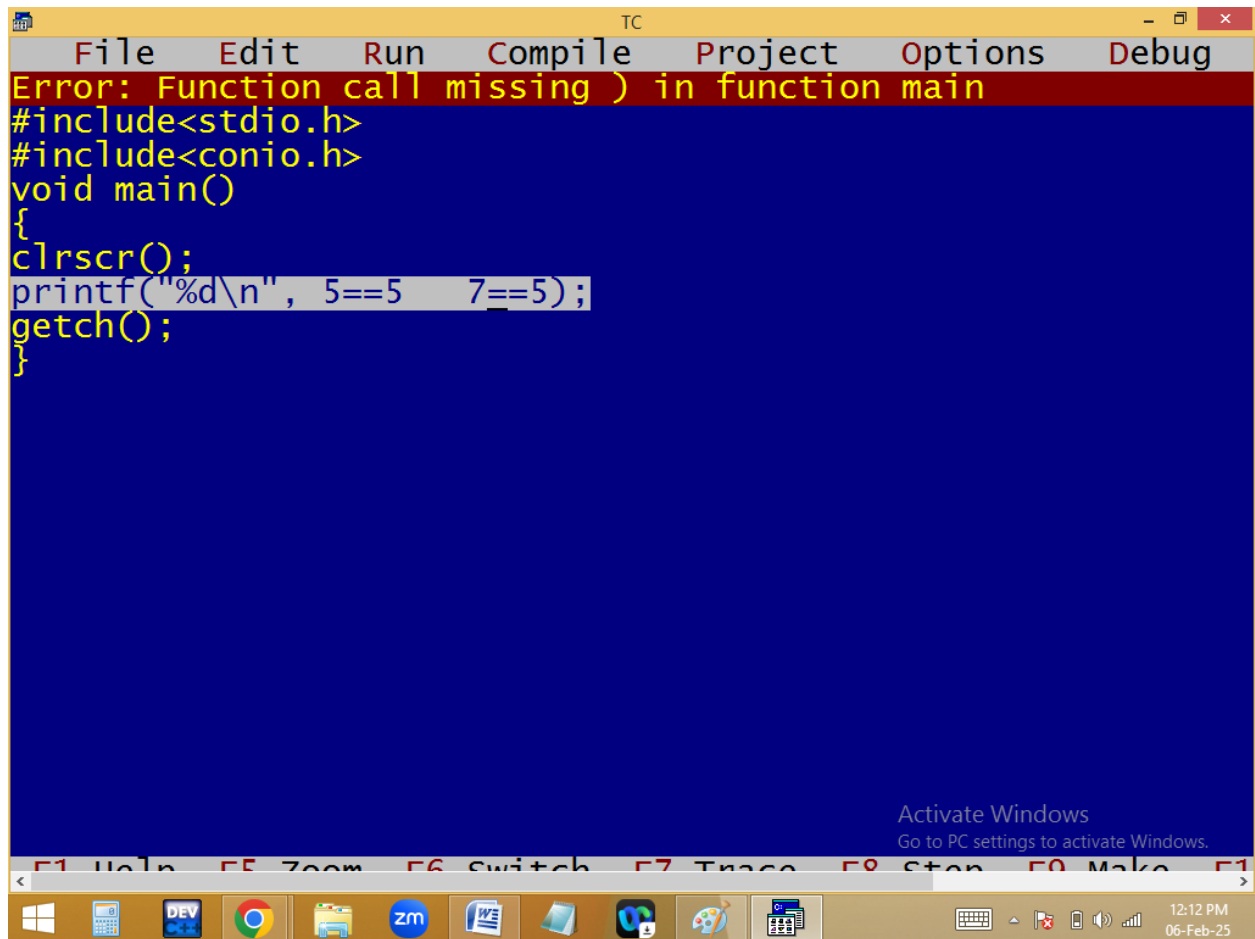
Note: In C && C++ 0 means false and other than 0 anything is true i.e. 1.

Truth tables:

Operator	Expression1	Expression2	Result
&&-and	1	1	1
	1	0	0
	0	1	0
	0	0	0
-or	1	1	1
	1	0	1
	0	1	1
	0	0	0

!true = false

!false = true



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top states: "Error: Function call missing) in function main". Below this, the source code is displayed on a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5 7==5);
getch();
}
```

The line `printf("%d\n", 5==5 7==5);` is highlighted with a light grey background. At the bottom of the IDE window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. An "Activate Windows" watermark is visible in the bottom right corner of the IDE area, with the text "Go to PC settings to activate Windows." The Windows taskbar at the very bottom shows various application icons (Windows Start, Task View, File Explorer, Chrome, Dev C++, Zoom, Word, etc.) and the system clock in the bottom right corner displays "12:12 PM" and "06-Feb-25".

TC

File Edit Run Compile Project Options Debug

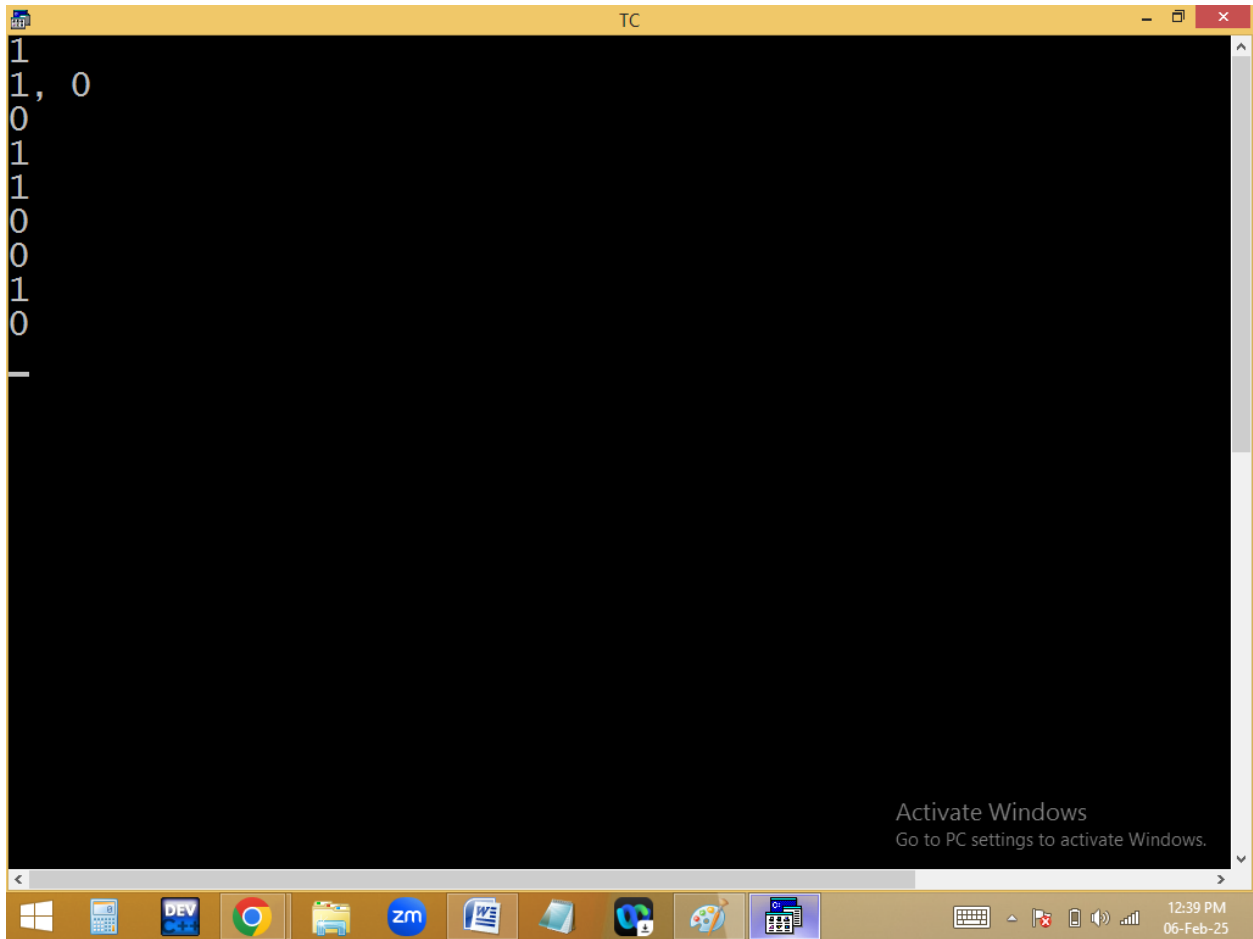
Line 14 Col 30 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5 , 7==5);
printf("%d, %d\n", 5==5 , 7==5);
printf("%d\n", 5==5 && 7==5);
printf("%d\n", 5==5 && 7!=5);
printf("%d\n", 5==5 || 7==5);
printf("%d\n", !(5>=5) );
printf("%d\n", 1==1 && 2<=2 && 3>3 );
printf("%d\n", 1==1 || 2<=2 && 3>3 );
printf("%d\n", (1==1 || 2<=2) && 3>3 );
getch();
}
```

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$$\begin{array}{c}
 \underline{5==5 \quad \&\& \quad 7!=5} \\
 \quad \&\& \quad \\
 \hline
 1
 \end{array}$$

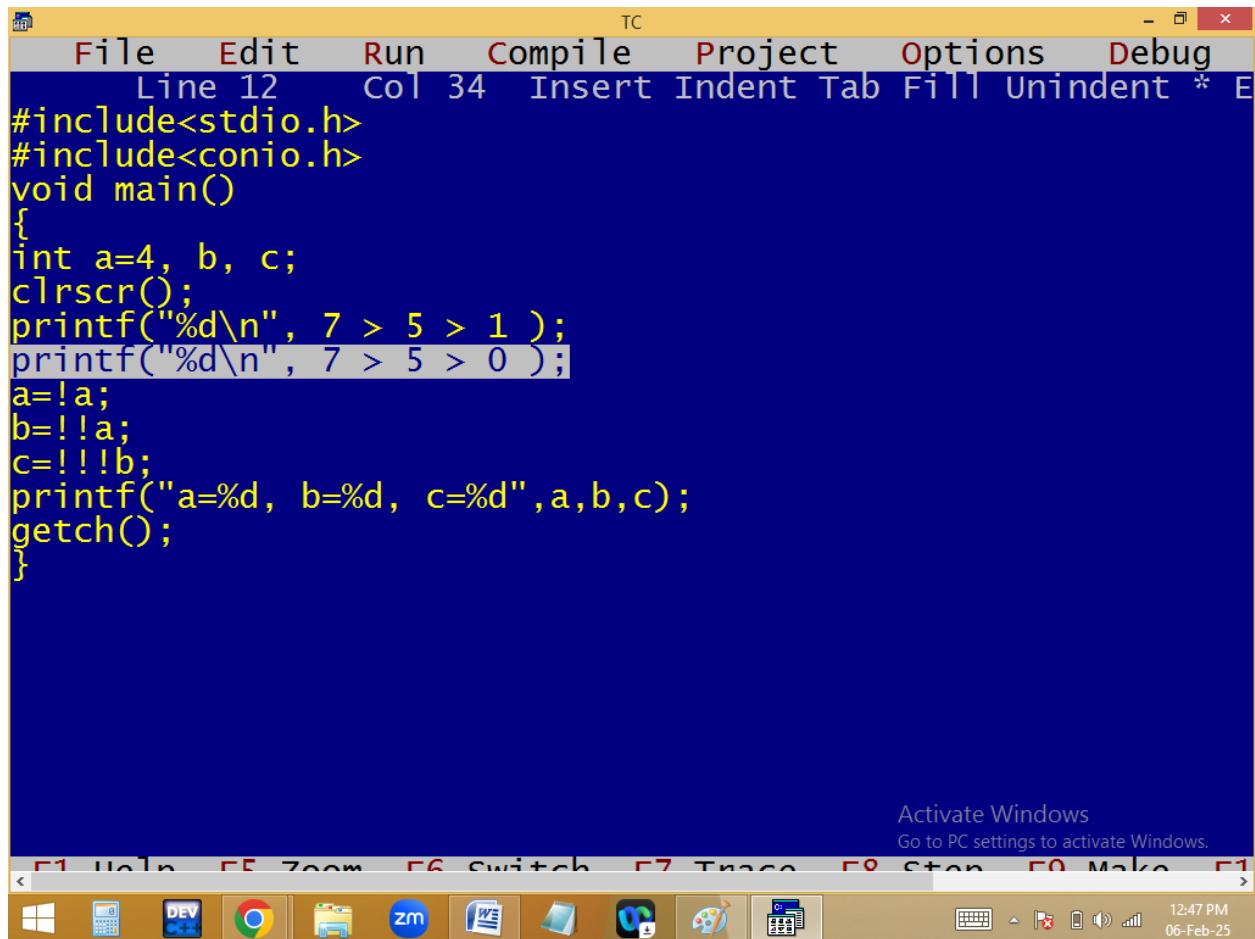
$$\begin{array}{c}
 \underline{1!=1 \quad || \quad 6==9} \\
 \quad 0 \quad || \quad 0 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{c}
 \underline{!(7>7)} \\
 \quad 0 \\
 \quad \downarrow \\
 \quad 1
 \end{array}$$

$$\begin{array}{c}
 \underline{1==1 \quad \&\& \quad 2<=2 \quad \&\& \quad 3>3} \\
 \quad 1 \quad \&\& \quad 1 \\
 \hline
 1 \quad \&\& \quad 0 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{c}
 \underline{(1==1 \quad || \quad 2<=2) \&\& \quad 3>3} \\
 \quad 1 \quad || \quad 1 \\
 \hline
 1 \quad \&\& \quad 0 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{c}
 \underline{1==1 \quad || \quad 2<=2 \quad \&\& \quad 3>3} \\
 \quad 1 \quad \&\& \quad 0 \\
 \hline
 1 \quad || \quad 0 \\
 \hline
 1
 \end{array}$$

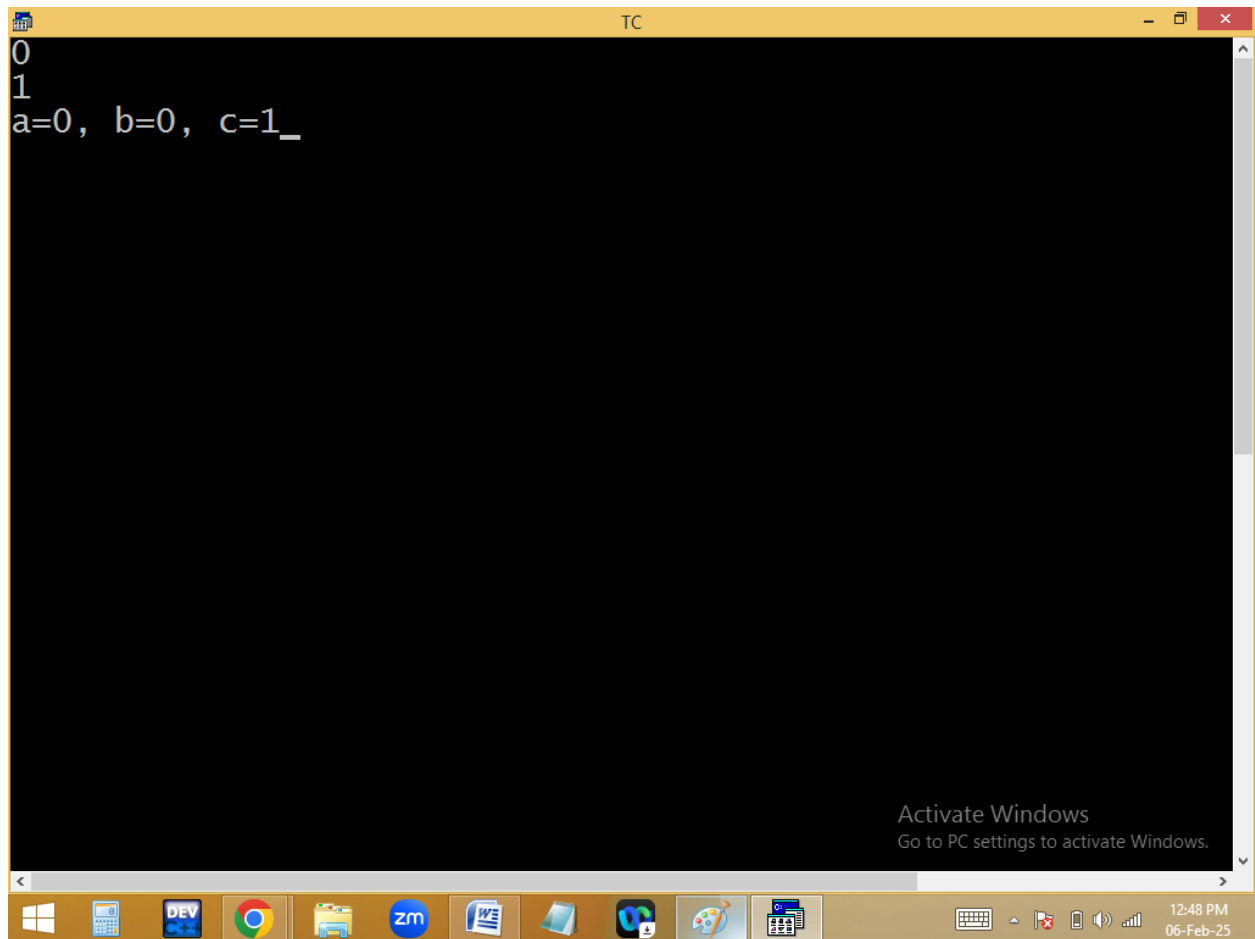


```
TC
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Line 12 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=4, b, c;
clrscr();
printf("%d\n", 7 > 5 > 1 );
printf("%d\n", 7 > 5 > 0 );
a=!a;
b=!a;
c=!b;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
```

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7 > 5 > 1

1 > 1
0

a=4

a = !a ==> !4 ==> !1 ==> a=0

b = !!a ==> !0 ==> 1 ==> !1 ==> b=0

c = !!!b ==> !0 ==> 1 ==> 0 ==> c=1

